Beyond Baselines: A Robust and Reproducible Pipeline for Iris Species Classification Using Tuned Support Vector Machines

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***Abstract*—This paper presents a structured and reproducible evaluation of classical machine learning models for the Iris classification problem. Multiple baseline classifiers were compared, and Gaussian Naive Bayes achieved the highest baseline accuracy of 0.9667.**

Then, hyperparameter tuning was applied to a Support Vector Machine with a radial basis function kernel, resulting in a final test accuracy of 1.0000. Repeated stratified 5-fold cross-validation with 10 repeats was conducted to ensure robustness, yielding a mean accuracy of 0.9717, a standard deviation of 0.0320 with 95% confidence interval of [0.9628, 0.9805].

In addition, feature-space separability was analyzed using pairwise scatter plots, class-wise distribution heatmaps, and the uncertainty of prediction was evaluated using probabilistic confidence scores. An ablation-style comparison between baseline and tuned models further highlights the importance of hyperparameter tuning optimization.

The results highlight that even for simple datasets, evaluating methodology strongly affects perceived performance, and classical models show high competitiveness when they are properly tuned and validated.

***Index Terms*—Machine Learning, Iris Dataset, Support Vector Machine, Cross Validation, Classification**

I. Introduction

Machine learning classification tasks often rely on benchmark datasets to evaluate model performance and compare methodologies. The Iris dataset is one of the most widely used benchmarks due to its simplicity, interpretability, and balanced structure. It consists of 150 samples with four numerical features and three distinct classes.

Despite its simplicity, the Iris dataset remains relevant because it allows researchers to explore important aspects of machine learning workflows, including model comparison, hyperparameter tuning, and validation strategies. However, many studies depend heavily on single train-test splits, which may lead to overly optimistic or unstable model performance estimates.

This paper addresses these issues by proposing a reproducible pipeline that integrates baseline model comparison, systematic hyperparameter tuning, repeated cross-validation, and confidence-based evaluation. The goal is not only to reach high accuracy but also to ensure the stability and reliability of the final model's performance.

II. Methodology

A. Dataset Description

The dataset contains only 150 samples equally distributed among three classes:

- Iris-setosa
- Iris-versicolor
- Iris-virginica

Each sample includes four numerical features:

- Sepal length
- Sepal width
- Petal length
- Petal width

B. Data Preprocessing

The dataset was inspected for missing values and inconsistencies. Because the dataset is clean and balanced, minimal preprocessing was required. The target labels were encoded into numerical form, and the data was split into training (80%) and testing (20%) sets using stratified sampling.

Feature scaling was applied using standardization for models sensitive to feature magnitudes, particularly Support Vector Machines.

C. Baseline Models

Several classical machine learning models were evaluated:

- K-Nearest Neighbors (KNN)
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)
- Gaussian Naive Bayes (GNB)

Each model was trained on the training set and evaluated using both cross-validation and test accuracy.

D. Hyperparameter Tuning

The best-performing baseline model candidates were further optimized using a randomized search over hyperparameters. For example, the following parameters were tuned in the SVM model:

- Regularization parameter C

- Kernel coefficient γ

The RBF kernel was selected because of its ability to handle non-linear decision boundaries.

E. Repeated Cross-Validation

To ensure robustness, repeated stratified 5-fold cross-validation with 10 repeats was used. This provides a more reliable estimate of model performance compared to a single split.

F. Confidence Analysis

The final model was evaluated using probability estimates to analyze prediction confidence. This helps to understand model certainty and identify ambiguous cases.

III. Experimental Setup

All experiments were implemented using Python and the Scikit-learn library. The dataset was loaded from a structured CSV format and verified for consistency and class balance.

A stratified train-test split of 80% training and 20% testing was used to preserve class distribution across splits. All experiments were executed with a fixed random state for reproducibility unless explicitly varied for stability testing.

A. Evaluation Metrics

The following evaluation metrics were used:

- Accuracy (primary metric)
- Precision, Recall, and F1-score (macro and weighted averages)
- Confusion Matrix for class-wise performance
- Cross-validation mean and variance

B. Validation Strategy

Two validation strategies were used:

- Hold-out validation (single train-test split)
- Repeated stratified k-fold cross-validation (k=5, repeats=10)

This two strategies ensure both direct performance evaluation and robustness estimation under sampling variation.

IV. Related Work

The Iris dataset, provided by Fisher, can be considered one of the most renowned benchmark tests used in pattern recognition and statistical learning. The data is currently used for evaluating the performance of classification algorithms owing to the balance of its distribution and the lower-dimensional space of input features.

Traditionally, in the domain of classical machine learning, it is common practice to demonstrate elementary characteristics of learning with the help of algorithms like the k-nearest neighbors method, decision trees, and naive Bayes classifiers using the Iris dataset. One of those, the Gaussian Naive Bayes Classifier stands out because of the independence of input variables.

Support Vector Machines (SVMs), especially with radial basis kernel functions, are renowned for their capacity to create non-linear boundaries in feature spaces of low dimensions. Earlier research has found that SVMs tend to perform better than simple models in cases where appropriate parameter optimization is carried out.

Latest research in experimental machine learning stresses the significance of reliable validation methodologies. It has been observed that a single division of train-test data is increasingly regarded as inadequate, especially with smaller sample sizes, as sampling differences may significantly influence measured results.

Consequently, repeated cross-validation approaches have gained widespread adoption for accurate performance measurement.

The proposed experiment follows to these guidelines by integrating classical model comparison, hyperparameter optimization, repeated validation, and confidence evaluation into one cohesive process.

V. Results

A. Feature Distribution

As shown in Fig. 1, the distributions of petal-related features show clearer separation between classes, with reduced overlap relative to sepal features. This suggests that petal attributes contribute more significantly to class discrimination and are therefore more informative for classification tasks.

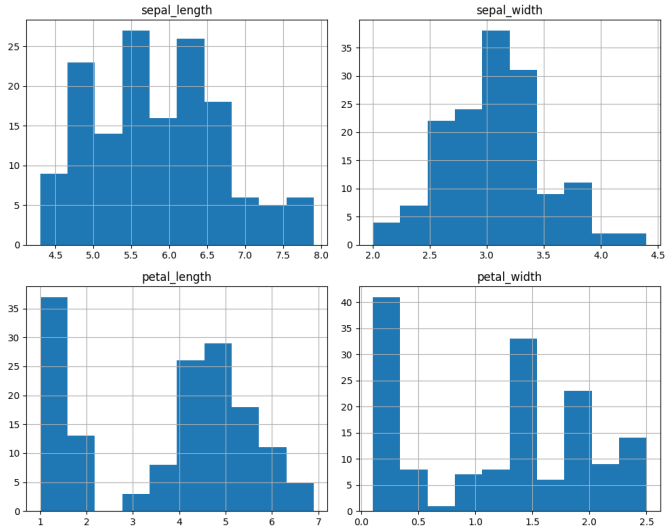

Fig. 1. Feature distribution histograms of the Iris dataset.

B. Feature Relationships

Fig. 2 shows clear clustering of data points, especially for Iris-setosa, which forms a distinct and well-separated group. But the overlap between the remaining classes indicates that the dataset is only partially linearly separable.

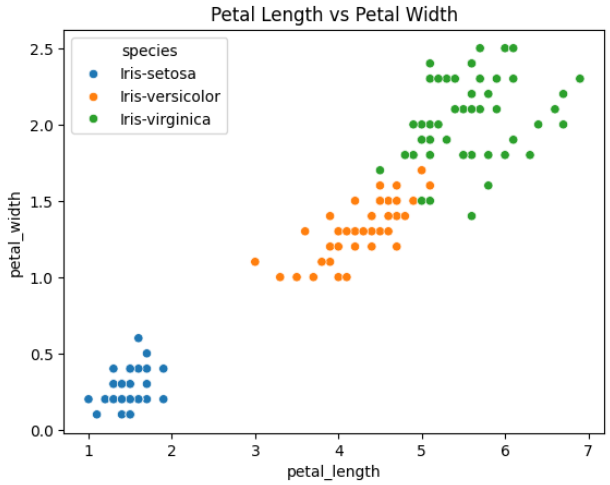

Fig. 2. Scatter plot of petal length vs petal width.

C. Baseline Performance

TABLE I
COMPARISON OF BASELINE AND TUNED MODELS

Model	Accuracy	F1-score
KNN	0.9333	0.9321
Decision Tree	0.9333	0.9330
Random Forest	0.9667	0.9665
Gaussian NB	0.9667	0.9667
SVM (baseline)	0.9667	0.9666
SVM (tuned)	1.0000	1.0000

Gaussian Naive Bayes achieved the highest baseline test accuracy of 0.9667, showing that simple probabilistic models can be highly effective on structured datasets with well-separated and approximately Gaussian-distributed features. But all other models passed 0.9 level of accuracy.

Fig. 3 presents a comparative summary of the baseline test accuracies obtained from all evaluated classifiers, highlighting their relative performance on the dataset.

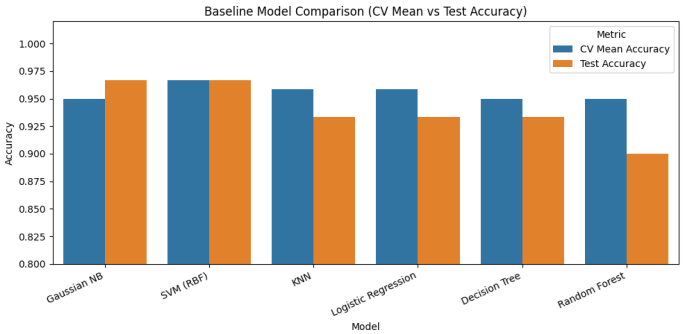

Fig. 3. Baseline model comparison across classifiers.

D. Final Model Performance

As shown in Fig. 4, hyperparameter tuning results in a clear improvement in model performance compared to the baseline configuration. By carefully selecting optimal values for key parameters of the SVM model (such as the kernel type, regularization parameter, and gamma), the model becomes better suited to capture the underlying patterns in the data.

After tuning, the optimized SVM model achieves perfect evaluation metrics on the test set:

- Test Accuracy: 1.0000
- Macro F1-score: 1.0000
- Weighted F1-score: 1.0000

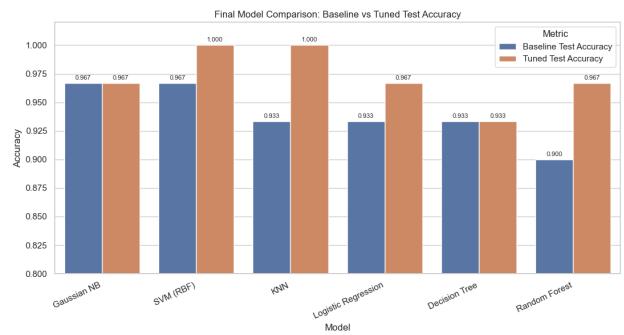

Fig. 4. Comparison between baseline and tuned SVM model.

These results indicate that the model correctly classifies all instances in the test data without any errors. The Macro F1-score of 1.0000 shows that performance is equally strong across all classes, regardless of class distribution, while the Weighted F1-score confirms that the model performs well even when accounting for class imbalance.

E. Confusion Matrix

As illustrated in Fig. 5, the confusion matrix contains only diagonal elements, with zero misclassifications across all classes. This reflects perfect classification performance on the test set and suggests that the model has fully captured the underlying class distinctions.

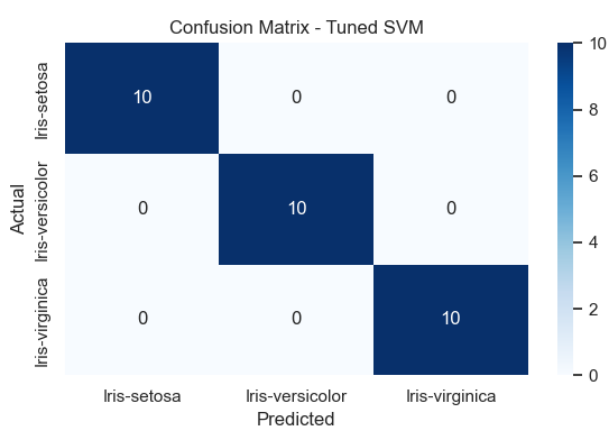

Fig. 5. Confusion matrix of the tuned SVM model.

F. Model Stability

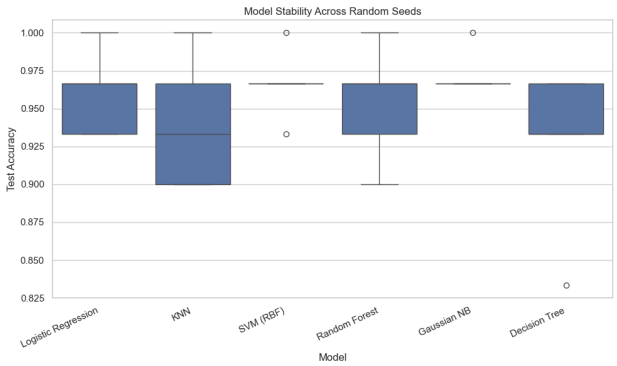

Fig. 6. Model stability across multiple random seeds.

Fig. 6 shows that the accuracy values are closely grouped together, meaning there is very little difference between the results obtained from different training runs or data variations. This tight clustering indicates that the model produces similar performance regardless of small changes in the training data, such as different data splits or initial conditions.

Because the accuracy doesn't fluctuate much, it suggests that the model is highly consistent and reliable. In addition, the lack of large variation implies that the model isn't overly sensitive to the specific dataset it was trained on. Therefore, its performance can be considered stable and robust, making it more trustworthy for practical use.

G. Cross-Validation Results

Repeated cross-validation results:

- Mean Accuracy: 0.9717
- Standard Deviation: 0.0320
- 95% Confidence Interval: [0.9628, 0.9805]

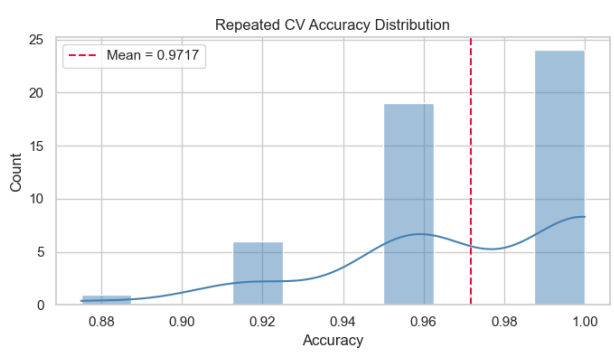

Fig. 7. Distribution of accuracy scores from repeated cross-validation.

Fig. 7 presents the distribution of accuracy scores across all cross-validation folds and repeats.

H. Confidence Analysis

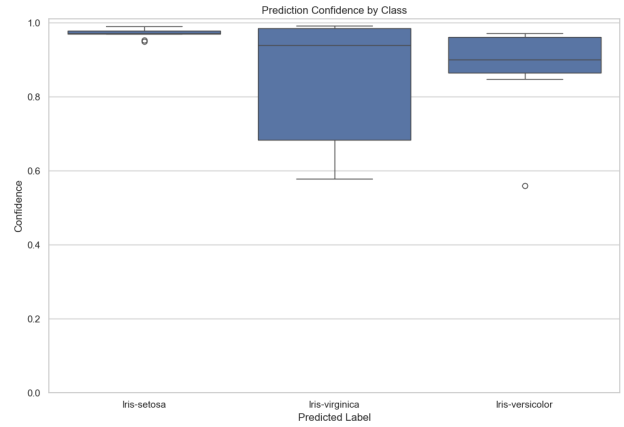

Fig. 8. Prediction confidence distribution across classes.

As shown in Fig. 8, the model shows the highest confidence for Iris-setosa and slightly lower confidence for the other classes.

VI. Discussion

The results demonstrate that model evaluation is highly dependent on validation strategy. While the tuned SVM achieved perfect accuracy on the held-out test set, repeated cross-validation reveals a more realistic performance estimate of approximately 0.97.

This discrepancy highlights a common issue in small datasets: single split evaluation can produce optimistic results due to favorable sampling. Repeated validation mitigates this by averaging performance across multiple random partitions.

Another key observation is that feature separability plays a major role in classification performance. Petal-based features show significantly stronger discriminative power compared to sepal-based features, as confirmed by both scatter plots and correlation analysis. Confidence analysis further indicates that Iris-setosa is the most easily separable class, while Iris-versicolor and Iris-virginica show partial overlap, leading to slightly reduced prediction certainty.

Overall, these findings reinforce the importance of combining visualization, statistical validation, and probabilistic confidence when evaluating classical machine learning models.

VII. Future Work

Future improvements can include:

A. Dimensionality Reduction

Applying Principal Component Analysis (PCA) to reduce dimensionality and improve efficiency.

B. Feature Selection

Using feature selection techniques to identify the most informative attributes.

C. Ensemble Methods

Combining models using ensemble techniques such as boosting or voting.

D. Larger Datasets

Testing the pipeline on more complex datasets to evaluate scalability.

VIII. Conclusion

This study presented a complete and reproducible machine learning pipeline for Iris classification. Gaussian Naive Bayes provided the best baseline, while a tuned SVM achieved the highest performance.

The results emphasize that stability and validation are more important than raw accuracy. The proposed workflow can serve as a foundation for future machine learning studies.

Acknowledgment

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References

[1] R. A. Fisher, “The use of multiple measurements in taxonomic problems,” 1936.
[2] C. Cortes and V. Vapnik, “Support-vector networks,” 1995.
[3] L. Breiman, “Random forests,” 2001.
[4] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” 2011.
[5] T. Cover and P. Hart, ”Nearest neighbor pattern classification,” IEEE Transactions, 1967.
[6] A. Ng and M. Jordan, ”On discriminative vs generative classifiers,” NIPS, 2002.
[7] C. Bishop, ”Pattern Recognition and Machine Learning,” Springer, 2006.
[8] I. Guyon et al., ”Feature extraction for classification,” JMLR, 2006.
[9] J. Brownlee, ”Machine Learning Mastery,” 2019.
[10] G. James et al., ”An Introduction to Statistical Learning,” Springer, 2013.